Zain Khan

Full-Stack Software Engineer | Applied ML & Edge AI

Calgary, AB · zain.mk@hotmail.com · (403) 615-9897
linkedin.com/in/zainmkhan · github.com/zainmk · zainkhan.vercel.app

SUMMARY

Full-stack software engineer with 5 years of production experience and a dual background in Mechatronics Systems Engineering and Computer Science. Shipped multi-tenant SaaS platforms, applied computer vision systems, and cloud infrastructure end-to-end — React frontends and Node.js APIs through Azure deployment, Terraform IaC, and CI/CD. Currently building edge AI on NVIDIA Jetson (TensorRT, YOLOv8, ROS 2), focused on deploying ML models to real hardware under real constraints.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL
AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, TensorRT, YOLOv8, computer vision, deep learning (CNN, GRU, RNN), Kalman filter, RAG, LLM integration, vector databases (ChromaDB), FastEmbed, sentence-transformers, K-Means clustering, NumPy, SciPy, Pandas
Full-Stack: ReactJS, Node.js, Express, FastAPI, Flask, GraphQL, REST APIs, Material UI, Knex, MySQL, PostgreSQL, MongoDB, Playwright, PyTest, PWA / Service Workers
Cloud & DevOps: Azure (App Service, MySQL Flexible Server, Static Web Apps, Key Vault, Blob Storage, Container Registry, App Configuration), Terraform (IaC), Docker, Kubernetes, GitHub Actions, CI/CD, RBAC, multi-tenant architecture
Data & ETL: PySpark, Databricks, ETL, data modeling, data visualization, Selenium
Embedded & Edge: NVIDIA Jetson Orin Nano, ROS 2, Raspberry Pi, Arduino, MATLAB, Simulink

WORK EXPERIENCE

GCT Maintenance Solutions — *Software Engineer* Calgary, AB | September 2024 – Present

- Lead full-stack development of a multi-tenant SaaS ERP platform for the mining industry using ReactJS, Node.js/Express, and MySQL (160-table schema).
- Designed and automated multi-tenant Azure architecture with Terraform (IaC), reducing environment deployment time from ~25–30 min to ~3 min.
- Built GitHub Actions CI/CD pipelines for frontend and backend across 6 concurrent Azure environments (2 production, dev, intern, and others).
- Reduced database recovery time from days to minutes via automated Azure backup and recovery plans; owned schema design and migration/seed workflows with Knex.
- Implemented Azure Key Vault secret management, App Configuration feature flags, RBAC-based authorization, and Blob Storage hot/cold retrieval tiers across all environments.
- Led ETL migration (SAP ERP → JDE) for 3 major Canadian mine sites using PySpark on Databricks — transforming and loading hundreds of tables and millions of records, with data auditing and client handover documentation.
- Led a team of 4 intern developers under Agile principles; defined software design coding standards and review practices.

Heliolytics — *Full-Stack Developer – Applied ML* Toronto, ON | August 2021 – June 2023

- Applied computer vision and ML algorithms for automated anomaly detection and classification in solar panel thermal images, improving analyst detection speed 5×.
- Led full-stack development of a ReactJS internal analysis web tool (migrated to TypeScript), replacing legacy tooling for a 30-person inspection team.
- Owned the model evaluation loop — data labeling and curation, thresholding, false-positive reduction, and inference integration — with PyTest coverage validating detection accuracy.
- Implemented K-Means clustering in backend endpoints for geopositional data subset isolation across hundreds of thousands of thermal images.
- Designed GraphQL schema around anomaly/asset-centric data models; scaled backend from tens to hundreds of RESTful CRUD endpoints.
- Built a Python-automated PDF report pipeline generating ~150 client anomaly reports per month with minimal errors.

- Developed streaming pipelines to ingest satellite imagery by geolocation for a SaaS solar array monitoring product.
- Reviewed 3–4 pull requests daily during high-growth phase; defined cross-functional workflows between product, design, and QA and translated CV design decisions for non-technical stakeholders.

PROJECTS

Jetson Edge AI Inspection Pipeline (in progress) | *Python, TensorRT, YOLOv8, ROS 2, NVIDIA Jetson Orin Nano* — Real-time visual inspection on edge hardware.

- Benchmarking TensorRT-optimized object detection across FP32/FP16/INT8 precision and power modes, with ROS 2 integration for downstream safety-zone monitoring.
- Documenting the full model-to-hardware path — quantization tradeoffs, latency budgets, and thermal constraints — as a public technical case study.

KalmanNET | *PyTorch, Flask, Three.js* — GRU-based KalmanNet vs. classical Kalman filter for drone sensor fusion under environmental stress.

- Implemented a GRU network (input=9 innovations, hidden=64, 4 per-sensor heads) trained on scripted calibration flight data with Adam optimizer and cosine LR schedule.
- Reduced drone position RMSE by 79.4% under barometer heat bias and 67.2% under combined sensor stress vs. the classical filter.
- Optimized training to ~32 s on CPU via 10× subsampling and vectorized batched GRU calls; deployed model at a ~70 KB footprint.

SensoRAG | *Python, Flask, ChromaDB, Claude API* — RAG-based sensor/transducer selection tool — upload datasheets, query in natural language.

- Built a Flask app ingesting PDF, TXT, CSV, and TSV datasheets; embedded documents with FastEmbed into a ChromaDB vector store.
- Integrated the Anthropic Claude API for RAG synthesis with contextual justification, combining hardware domain knowledge with LLM retrieval.

autoadgen | *GitHub Actions, Claude API, Playwright, Jinja2* — Automated weekly marketing banner pipeline for Shopify stores.

- Built a scheduled CI/CD pipeline generating 3 LLM-produced ad copy concepts per run, reducing copywriting from 1–2 hrs/week to ~2 seconds.
- Rendered 1920×640 px banners with Playwright/Chromium from Jinja2 templates, cutting design/layout from 2–4 hrs/week to ~30 seconds; integrated Shopify GraphQL API and a human-in-the-loop approval gate.

vectorize | *Python, ChromaDB, sentence-transformers* — Semantic search explorer comparing BM25 keyword search vs. vector embeddings.

- Implemented BM25 and ChromaDB semantic search side-by-side (distilled model2vec static model to fit Vercel's 500 MB limit); documented contextual vs. static embedding tradeoffs.

htmlr | *React 19, TypeScript, TipTap, PWA* — Offline-first WYSIWYG note editor storing notes as self-contained .html files locally.

- Built with the File System Access API for direct folder read/write and an IndexedDB cache layer; service worker via vite-plugin-pwa with timestamp-based conflict resolution for network-mount sync.

EDUCATION

Western University — *Dual Degree* London, ON | 2016 – 2021

- Bachelor of Engineering Science (BESc), Mechatronics Systems Engineering
- Bachelor of Science (BSc), Computer Science

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — February 2025
- **IBM AI Engineering – Coursera** — ML, deep learning, CNNs, RNNs, scikit-learn, PyTorch, TensorFlow, Apache Spark; August 2024